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PERIIOPERATIVE

Peri-operative analgesic management of patients at increased risk of compartment syndrome

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Key Points

- Acute Compartment Syndrome (ACS) is a potentially life-changing complication following trauma, but fear of missing a diagnosis should not limit the quality of analgesia offered.
- Adequate analgesia should be offered to every patient, with full consent for the risks and benefits of each option.
- A multimodal analgesic approach should always be adopted.
- Where appropriate, a nerve block may be utilised. A low-concentration local anaesthetic choice should be used in most patients.
- Adequate vigilance is the key to diagnosis of ACS, and includes adequate clinical monitoring, patient education and involvement, and use of compartment pressure measurement.
- Treatment for suspected ACS is urgent, therefore rapid senior surgical review is required.

Background

Compartment syndrome is a rare but devastating condition, requiring high vigilance for diagnosis, and rapid management to avoid potential extensive tissue loss.

Adequate analgesia is essential for all trauma patients. Multi modal analgesia should be tailored to each patient, with full consent for the risks and benefits of each option.

Fear of masking compartment syndrome should not be a reason to avoid regional nerve blockade if it is indicated. Varying forms of analgesia have been blamed in case reports for delaying or missing a diagnosis of compartment syndrome. There is no evidence that regional nerve blockade produces a higher risk than other options, including opiate PCAs. Regional nerve blockade is more effective at blocking the myelinated nerve cells carrying somatic and visceral pain, than the unmyelinated C fibres which carry signals initiated by ischaemic pain. Regional nerve blockade may help with diagnosis as breakthrough pain despite a nerve blockade is considered a red flag.

Adequate vigilance is the key to diagnosis, and includes adequate clinical monitoring, patient education and involvement, and use of compartment pressure measurement. Pain is the most common symptom, but is not present in all cases of compartment syndrome, therefore a high degree of clinical suspicion is required.

Treatment of suspected compartment syndrome is urgent. Fasciotomies should be performed within 4-6 hours. Rapid senior surgical review is therefore required.

This guideline is based on consensus opinion, as discussed in the Association of Anaesthetists Guideline, 'Regional analgesia for lower leg trauma and the risk of acute compartment syndrome.' There is a lack of high quality trials, therefore this opinion is based on case reports and small case series.

Risk factors for compartment syndrome

Type of injury:

- Tibial shaft fractures
- Distal radius fractures
- Diaphyseal forearm fracture
- Femoral diaphyseal fracture
- Tibial plateau fracture
- Crush injuries
- Open fractures
- Vascular injuries
- High energy injury

Patient factors:

- Male sex
- Anticoagulation
- Haemophilia

Surgical factors:

- Intramedullary nailing
- Use of tourniquets

Analgesic Management

Decisions regarding analgesia management should be jointly made between the surgical team and the anaesthetist. In rare circumstances where the surgeon and anaesthetist cannot reach an agreement it should be remembered that the patient always has the final say, provided they have been consented with all the risks, benefits and alternatives regarding analgesic options.

To directly quote the Association of Anaesthetists:

"Although the patient has the right to refuse any form of treatment, such as the analgesic technique offered or the surgical procedure proposed, neither the surgeon nor the anaesthetist has the right to veto a treatment recommended by the other. Ideally, consensus should be achieved but, if consensus is not achievable, the role of the anaesthetist as the expert on pain relief should be respected. It is, therefore, the anaesthetist who has the right to offer the patient the range of what they consider to be acceptable analgesic techniques provided they express to the patient the concerns voiced by the surgeon."

A multimodal approach should always be adopted, and a regional block is considered as part of the multi-modal approach.

With regards to a regional nerve blockade, the anaesthetist should consider:

- A low-density, analgesic block in patients at risk of ACS. Options include 0.125-0.25% Chirocaine or 0.25% Ropivicaine. Adjuncts within the regional nerve block should be avoided.

- The exception to the low-dose rule would be patients who would benefit from having their procedure performed solely under regional nerve blockade. For example, patients with underlying significant co-morbidities, placing them at high risk for a general anaesthetic.
- Of note, there has been no case reports of sciatic, obturator or popliteal nerve blocks causing ACS.

In cases where the anaesthetist is unable to perform a regional technique as part of the multi-modal approach, a high dose opioid spinal should be considered as an alternative for lower limb trauma. There have been no case reports of spinal anaesthesia masking ACS.

Post-operative analgesia via the epidural route should be avoided as case reports have implicated epidurals with delayed diagnosis of ACS.

Dense, long duration blocks (both peripheral or neuraxial) which last significantly longer than the surgery, should be avoided.

References

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